

Automated Dynamic Analysis Signature Generation

Using LLMs for Malware Behavior Understanding and CAPEv2 Signature Creation

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March 22, 2026

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Manual workflow today

- ▶ Signature writing is slow
- ▶ Inconsistent quality between analysts
- ▶ Hard to scale to large malware volumes
- ▶ Repetitive effort for similar families

Target workflow

- ▶ AI-assisted signature drafting
- ▶ Faster analyst turnaround
- ▶ Standardized detection logic
- ▶ Human validation before deployment

Core question: Can LLMs generate useful CAPEv2 signatures with strong precision and practical analyst value?

- ▶ Dataset: 100 malware samples across 10 families
- ▶ Input: CAPEv2 dynamic analysis JSON reports
- ▶ Output: Python CAPEv2 signatures + MITRE ATT&CK mappings
- ▶ Constraint: Low-cost approach using local Llama inference

Deliverables

6 notebooks, processed outputs, 20 generated signatures, evaluation report, and containerized setup.

1. Data Acquisition

2. Feature Extraction

3. Behavior IR

4. LLM Analysis

5. Signature Generation

6. Evaluation

Design principle

Separate data normalization from LLM reasoning to improve reliability and reproducibility.

Extracted behavior groups

- ▶ Process and injection activity
- ▶ Registry persistence indicators
- ▶ File operations and drops
- ▶ Network and command behavior

Behavior IR value

- ▶ Normalizes inconsistent report structures
- ▶ Reduces prompt noise for the LLM
- ▶ Improves reproducibility across runs

- ▶ Llama used for semantic interpretation (local, low cost)
- ▶ Family-level aggregation to detect recurring patterns
- ▶ CAPEv2 signature generation with strict validation:
 - syntax checks
 - logic checks
 - manual analyst review

Important result: Validation caught approximately 15% of generated signature issues before final output.

Challenge	How we solved it
Inconsistent JSON structures	Built robust parsing + Behavior IR abstraction
LLM output quality drift	Added cache, tighter prompts, post-checks
Signature correctness gaps	Added syntax + logic + manual validation
Lower ATT&CK edge-case accuracy	Added confidence-aware review workflow

- ▶ Feeding raw CAPEv2 JSON directly to Llama
- ▶ Fully automated ATT&CK mapping without confidence checks
- ▶ Treating generated Python as deployment-ready without validation

Key lesson

Automation in security must include explicit validation gates. Speed without verification creates detection risk.

Precision

86%

Recall

79%

F1-score

0.82

Quality Details

- ▶ False positive rate: 8%
- ▶ Validation catches edge-case bugs
- ▶ Human review remains in the loop

Output Volume

- ▶ 20 CAPEv2 signatures
- ▶ 10 malware families covered
- ▶ 35+ ATT&CK techniques mapped

- ▶ Cross-family precision on unseen families: 77%
- ▶ Cross-family recall on unseen families: 65%
- ▶ ATT&CK mapping overall accuracy: 84%

Interpretation

The model captures reusable behavior patterns, but still requires analyst oversight for edge cases such as subtle Discovery and Lateral Movement indicators.

- ▶ Practical analyst acceleration: from hours to minutes for initial signature drafts
- ▶ Repeatable pipeline with containerized setup
- ▶ Cost-aware execution using local inference
- ▶ Strong baseline for SOC workflow augmentation

Bottom line: this is an analyst multiplier, not a full analyst replacement.

- ① LLM-assisted CAPEv2 signature generation is feasible and useful.
- ② A Behavior IR layer is critical for stability and quality.
- ③ Validation is mandatory for safe security automation.
- ④ The system provides immediate practical value with clear known limits.

- ▶ Expand dataset size and family diversity
- ▶ Improve low-confidence ATT&CK mapping with hybrid rules
- ▶ Add automated regression tests for signatures
- ▶ Integrate with SOC tooling for continuous feedback

Thank you

Questions?